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Unit – 4 : Session – 12 : SLO – 1 | Influence on Instruction Sets — Choose the Best Answer

■ I. Multiple Choice

1. b) Simplified, single-cycle instructions — RISC design is driven by simple instructions that execute in one cycle.
2. a) Easier pipelining and decoding — fixed-length instructions align fetch/decode stages neatly.
3. c) Memory-to-memory operations — CISC instructions can operate directly on memory operands.
4. b) SIMD — one instruction processing multiple data elements boosts modern performance.
5. a) Code density — embedded systems have little memory, so compact code matters most.
6. a) To reduce branch penalties — predication executes both paths conditionally, eliminating branches.
7. b) Backward compatibility — x86 kept variable-length instructions to run legacy software.
8. c) More compact code — register-memory instructions do more per instruction, shrinking code size.
9. b) Out-of-order execution — dynamic scheduling most reshaped modern ISA/microarchitecture design.
10. a) Increased compiler complexity vs. code flexibility — more addressing modes give flexibility but complicate the compiler and hardware.